DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
DBMS MINI-PROJECT EVIDENCE REPORT

SMARTRETAIL
AI-POWERED INVENTORY AND SALES MANAGEMENT SYSTEM

DATABASE MANAGEMENT SYSTEMS (DA2)
COMPLETE SQL & PL/SQL IMPLEMENTATION EVIDENCE

Target RDBMS Engine:	Oracle Database 11g Express Edition Release 11.2.0.2.0 (64-bit)
Target Schema User:	SYSTEM (Verified Local Project Account)
Implementation Scope:	19 Approved Relations (Normalized Relational Model)
Primary Source of Truth:	SmartRetail_DA2_Antigravity_Handoff.docx
Schema Conformance Audit:	19/19 Relations Verified (SCHEMA_CONFORMANCE_AUDIT.md)
Test Suite Validation:	13/13 Automated Tests PASSED (run_all.log)

ACADEMIC YEAR 2025–2026 | DIGITAL ASSIGNMENT 2
STRICT SCHEMA CONFORMANCE AND EXECUTION VERIFICATION

1. System Overview & Architecture

SmartRetail is a normalized relational database management system designed to support modern supermarket retail operations. The architecture captures the complete product lifecycle from supplier procurement and warehouse receiving to point-of-sale customer checkout and inventory movement.

The operational workflow is divided into four functional sub-systems:

1. Procurement & Supplier Relations: Tracks suppliers, communication channels, purchase orders, purchase order line extensions, shipments, and receiving records.

2. Catalog & Merchandising: Organizes product categories and products with unit prices, barcodes, units of measure, and active merchandising flags.

3. Multi-Tier Inventory Management: Separates warehouse bulk storage locations from retail display shelves. Tracks stock balances, reorder thresholds, immutable audit movements in InventoryTransaction, and reactive restock and reorder events.

4. Point of Sale (POS) Checkout & Billing: Records transaction receipts, sales transaction lines, cashier attribution, split payment settlement across cash, card, and UPI, and customer loyalty records.

2. Approved Relational Schema Verification

The implemented database contains exactly 19 approved relations, adhering strictly to the architecture brief defined in SmartRetail_DA2_Antigravity_Handoff.docx. A formal Schema Conformance Audit verified 100% attribute, primary key, foreign key, and check constraint parity against the live Oracle 11g database.

Relation Name	Attributes / Columns	Integrity Constraints	Functional Purpose
Supplier	supplier_id, supplier_name, contact_person, email, gst_number, street, city, state, pincode, rating, created_at	PK: supplier_id UK: email, gst_number CK: rating BETWEEN 1 AND 5 NN: supplier_name, email, gst_number	Master vendor registry with commercial, GST tax registration, postal location, and vendor rating attributes.
SupplierPhone	supplier_id, phone_number	PK: (supplier_id, phone_number) FK: supplier_id -> Supplier(supplier_id) NN: phone_number	Multivalued 1:N vendor contact telephone numbers.
ProductCategory	category_id, category_name, parent_category_id	PK: category_id UK: category_name FK: parent_category_id -> ProductCategory(category_id) NN: category_name	Hierarchical merchandise taxonomy with self-referencing subcategory classification.
Product	product_id, sku, barcode, product_name, description, category_id, unit_of_measure, standard_unit_price, is_active	PK: product_id UK: sku, barcode FK: category_id -> ProductCategory(category_id) CK: standard_unit_price > 0, is_active IN ('TRUE','FALSE') NN: sku, barcode, product_name, category_id, unit_of_measure, standard_unit_price, is_active	Master merchandise catalog with retail pricing, barcode tracking, and active status.

Relation Name	Attributes / Columns	Integrity Constraints	Functional Purpose
Employee	employee_id, full_name, role, email, phone, hire_date	PK: employee_id UK: email CK: role IN ('MANAGER','CASHIER','INVENTORY_CLERK','ADMIN') NN: full_name, role, email, hire_date	Store personnel directory with defined operational role-based access segregation.
PurchaseOrder	po_id, po_number, supplier_id, created_by, order_date, expected_delivery_date, status, total_amount, is_auto_generated	PK: po_id UK: po_number FK: supplier_id -> Supplier(supplier_id), created_by -> Employee(employee_id) CK: status IN ('PENDING','APPROVED','PARTIALLY_RECEIVED','RECEIVED','CANCELLED'), is_auto_generated IN ('TRUE','FALSE'), total_amount >= 0 NN: po_number, supplier_id, created_by, order_date, status, total_amount, is_auto_generated	Procurement purchase order headers placed with certified commercial suppliers.
PurchaseOrderLine	po_id, product_id, quantity_ordered, unit_price	PK: (po_id, product_id) FK: po_id -> PurchaseOrder(po_id), product_id -> Product(product_id) CK: quantity_ordered > 0, unit_price > 0 NN: quantity_ordered, unit_price	Itemized procurement order lines and contracted procurement unit costs.
Shipment	shipment_id, tracking_number, po_id, carrier, shipment_date, received_date, status, received_by	PK: shipment_id UK: tracking_number FK: po_id -> PurchaseOrder(po_id), received_by -> Employee(employee_id) CK: status IN ('IN_TRANSIT','DELIVERED','CANCELLED') NN: tracking_number, po_id, carrier, shipment_date, status	Inbound supplier logistics and warehouse receiving tracking.
ShipmentLine	shipment_id, product_id, quantity_shipped, quantity_received	PK: (shipment_id, product_id) FK: shipment_id -> Shipment(shipment_id), product_id -> Product(product_id) CK: quantity_shipped >= 0, quantity_received >= 0 NN: quantity_shipped, quantity_received	Physical receiving reconciliation supporting partial delivery verification.
InventoryLocation	location_id, location_code, location_type, location_name, parent_location_id, capacity, aisle_number, zone	PK: location_id UK: location_code FK: parent_location_id -> InventoryLocation(location_id) CK: location_type IN ('WAREHOUSE','SHELF') NN: location_code, location_type, location_name	Hierarchical physical storage mapping distinguishing warehouses from display shelves.
InventoryStock	location_id, product_id, quantity_on_hand, reorder_threshold, reorder_quantity, last_updated_at	PK: (location_id, product_id) FK: location_id -> InventoryLocation(location_id), product_id -> Product(product_id) CK: quantity_on_hand >= 0, reorder_threshold >= 0, reorder_quantity > 0 NN: quantity_on_hand, reorder_threshold, reorder_quantity, last_updated_at	Live stock balance ledger per location with automated replenishment thresholds.
InventoryTransaction	transaction_id, product_id, location_id, transaction_type, quantity, transaction_date, reference_type, reference_id	PK: transaction_id FK: product_id -> Product(product_id), location_id -> InventoryLocation(location_id) CK: transaction_type IN ('RECEIPT','TRANSFER_IN','TRANSFER_OUT','SALE','ADJUSTMENT'), reference_type IN ('SHIPMENT','RESTOCK','SALE','MANUAL'), quantity > 0 NN: product_id, location_id, transaction_type, quantity, transaction_date	Immutable audit ledger recording all physical stock additions and deductions.

Relation Name	Attributes / Columns	Integrity Constraints	Functional Purpose
RestockEvent	restock_id, product_id, from_location_id, to_location_id, quantity_moved, event_date, triggered_by, performed_by, status	PK: restock_id FK: product_id -> Product(product_id), from_location_id -> InventoryLocation(location_id), to_location_id -> InventoryLocation(location_id), performed_by -> Employee(employee_id) CK: status IN ('PENDING','COMPLETED'), quantity_moved > 0 NN: product_id, from_location_id, to_location_id, quantity_moved, event_date, triggered_by, status	Warehouse-to-shelf internal replenishment tracking for depleted retail shelves.
ReorderTrigger	trigger_id, product_id, location_id, stock_at_trigger, threshold_at_trigger, trigger_date, generated_po_id, status	PK: trigger_id FK: product_id -> Product(product_id), location_id -> InventoryLocation(location_id), generated_po_id -> PurchaseOrder(po_id) CK: status IN ('OPEN','PO_CREATED','RESOLVED','IGNORED') NN: product_id, location_id, stock_at_trigger, threshold_at_trigger, trigger_date, status	Automated procurement reorder triggers logged when warehouse stock is breached.
Customer	customer_id, first_name, last_name, email, loyalty_points, membership_type, registered_date	PK: customer_id UK: email CK: loyalty_points >= 0, membership_type IN ('REGULAR','SILVER','GOLD','PLATINUM') NN: first_name, last_name, loyalty_points, membership_type, registered_date	Retail customer registry with loyalty point balances and tier profiles.
CustomerPhone	customer_id, phone_number	PK: (customer_id, phone_number) FK: customer_id -> Customer(customer_id) NN: phone_number	Multivalued 1:N customer contact telephone numbers.
SalesTransaction	transaction_id, receipt_number, customer_id, cashier_id, transaction_date, total_amount, status	PK: transaction_id UK: receipt_number FK: customer_id -> Customer(customer_id), cashier_id -> Employee(employee_id) CK: status IN ('COMPLETED','VOID','REFUNDED'), total_amount >= 0 NN: receipt_number, cashier_id, transaction_date, total_amount, status	Point-of-sale checkout invoice headers supporting guest customers.
SalesTransactionLine	transaction_id, product_id, shelf_location_id, quantity_sold, unit_price	PK: (transaction_id, product_id) FK: transaction_id -> SalesTransaction(transaction_id), product_id -> Product(product_id), shelf_location_id -> InventoryLocation(location_id) CK: quantity_sold > 0, unit_price > 0 NN: shelf_location_id, quantity_sold, unit_price	Itemized POS sale lines recording sold unit price and source shelf location.
Payment	payment_id, transaction_id, payment_method, amount, payment_date, reference_number, status	PK: payment_id FK: transaction_id -> SalesTransaction(transaction_id) CK: payment_method IN ('CASH','CREDIT_CARD','DEBIT_CARD','UPI','WALLET','POINTS'), status IN ('SUCCESS','PENDING','FAILED','REFUNDED'), amount > 0 NN: transaction_id, payment_method, amount, payment_date, status	Payment transactions supporting multi-tender split settlement across methods.

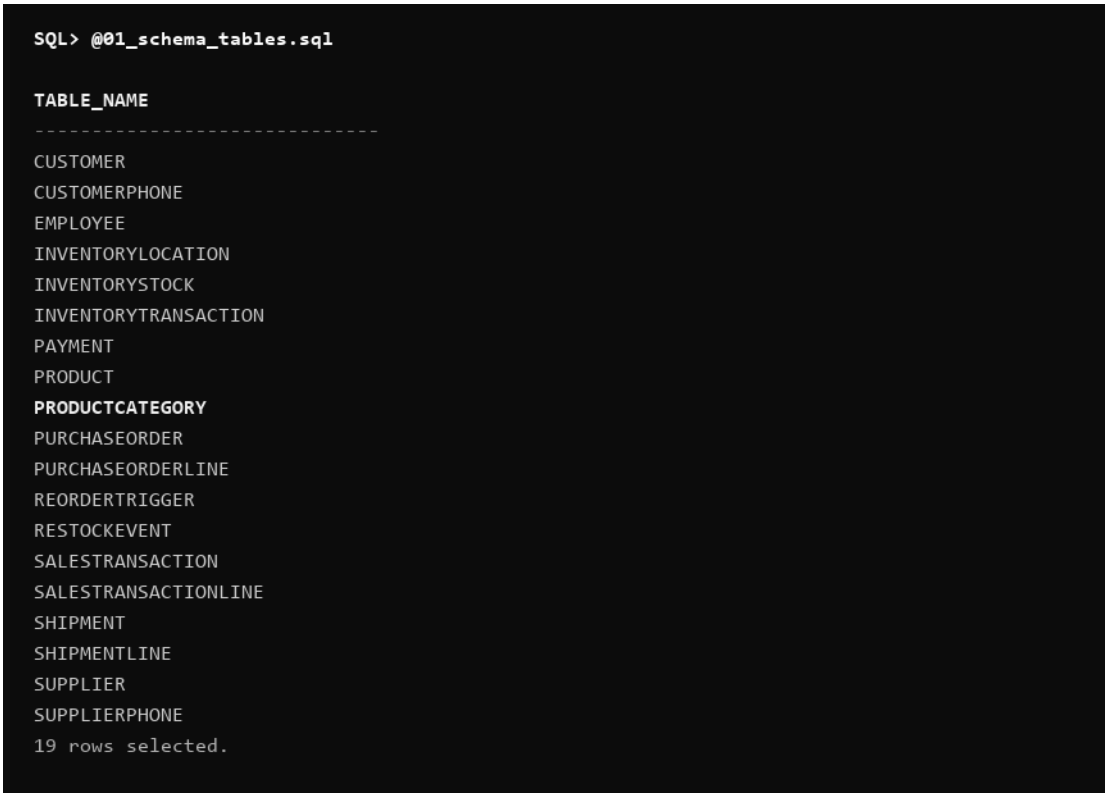


*Figure 1 — SmartRetail 19-Relation Schema Verification (SQL*Plus Query on USER_TABLES)*

3. Data Population

A coherent supermarket seed dataset was populated across all 19 tables in sql/03_sample_data.sql. The catalog models realistic consumer goods spanning Dairy, Bakery, Beverages, Personal Care, and Pantry Staples.

Every relation contains non-empty, validated data with verified referential integrity, supporting partial shipments, split tender payments, low shelf stock threshold scenarios, and warehouse reorder triggers.

```
SQL> @02_product_data.sql

   ID SKU          BARCODE        PRODUCT_NAME                 CAT UOM          PRICE ACTIVE
----- ------------ -------------- ---------------------------- ----- ---------- -------- ------
    1 SKU-DRY-001  8901030010011  Heritage Full Cream Milk 1L      6 LITER         64.00 TRUE
    2 SKU-DRY-002  8901030010028  Amul Salted Butter 500g          7 PACK         275.00 TRUE
    3 SKU-DRY-003  8901030010035  Amul Processed Cheese Blocks     7 PACK         310.00 TRUE
    4 SKU-BAK-001  8901030020010  Britannia Whole Wheat Bread      8 PACK          50.00 TRUE
    5 SKU-BAK-002  8901030020027  Britannia Good Day Butter Co     9 PACK          45.00 TRUE
    6 SKU-BAK-003  8901030020034  Bourbon Chocolate Cream Bisc     9 PACK          35.00 TRUE
    7 SKU-BEV-001  8901030030019  Real Fruit Power Mixed Fruit    10 TETRAPACK    120.00 TRUE
    8 SKU-BEV-002  8901030030026  Tropicana 100 Orange Juice 1    10 TETRAPACK    145.00 TRUE
    9 SKU-BEV-003  8901030030033  Tata Tea Gold Leaf 500g         11 PACK         320.00 TRUE
   10 SKU-BEV-004  8901030030040  Nescafe Classic Instant Coff    11 JAR          460.00 TRUE
   11 SKU-PER-001  8901030040018  Godrej Cinthol Original Soap     4 PACK         140.00 TRUE
   12 SKU-PER-002  8901030040025  Godrej Protekt Germ Liquid H     4 BOTTLE        99.00 TRUE
   13 SKU-STP-001  8901030050017  Organic Harvest Sona Masoori     5 BAG          390.00 TRUE
   14 SKU-STP-002  8901030050024  Organic Harvest Whole Sharba     5 BAG          285.00 TRUE
   15 SKU-STP-003  8901030050031  Organic Toor Dal Unpolished      5 PACK         175.00 TRUE
15 rows selected.
```

*Figure 2 — Product Relation Sample Data (Clean SQL*Plus Formatted Column Output)*

```
SQL> @03_inventory_data.sql

LOC_CODE       LOC_TYPE   SKU          PRODUCT_NAME                 QOH THRESH REORD_QTY
-------------- ---------- ------------ ---------------------------- ----- ------ ---------
WH-CENTRAL-01  WAREHOUSE  SKU-BAK-001  Britannia Whole Wheat Bread    120     30        80
WH-COLD-01     WAREHOUSE  SKU-DRY-001  Heritage Full Cream Milk 1L    180     50       100
WH-COLD-01     WAREHOUSE  SKU-DRY-002  Amul Salted Butter 500g         30     10        25
WH-COLD-01     WAREHOUSE  SKU-DRY-003  Amul Processed Cheese Blocks    18      5        15
SH-DAIRY-01    SHELF      SKU-DRY-001  Heritage Full Cream Milk 1L      8     15        30
SH-DAIRY-01    SHELF      SKU-DRY-002  Amul Salted Butter 500g         16      5        15
SH-DAIRY-01    SHELF      SKU-DRY-003  Amul Processed Cheese Blocks    10      4        10
SH-BAKERY-01   SHELF      SKU-BAK-001  Britannia Whole Wheat Bread      5     10        40
8 rows selected.
```

Figure 3 — Multi-Location Inventory Balances (JOIN InventoryLocation and InventoryStock)

4. SQL Query Demonstrations

4.1 Multi-Table JOIN Demonstration

The following query demonstrates a 5-table relational join linking SalesTransaction, SalesTransactionLine, Product, ProductCategory, and InventoryLocation (shelf). It computes itemized line extensions and attributes each line to the operating cashier.

```
SQL> @04_multi_table_join.sql

RECEIPT_NO        DATE       CASHIER              PRODUCT                  CATEGORY         SHELF          QTY    PRICE     TOTAL
----------------- ---------- -------------------- ------------------------ ---------------- -------------- ---- -------- ---------
RCP-2026-0001     2026-02-15 Divya Nair           Heritage Full Cream Milk Fresh Milk & Cre SH-DAIRY-01       2    64.00    128.00
RCP-2026-0001     2026-02-15 Divya Nair           Amul Salted Butter 500g  Butter & Cheese  SH-DAIRY-01       1   275.00    275.00
RCP-2026-0002     2026-02-15 Karthik Subramanian  Britannia Whole Wheat Br Artisan Breads   SH-BAKERY-01      1    50.00     50.00
RCP-2026-0003     2026-02-16 Divya Nair           Organic Harvest Sona Mas Staples & Grains SH-STAPLES-01     1   390.00    390.00
RCP-TEST-PROC-01  2026-09-17 Divya Nair           Amul Salted Butter 500g  Butter & Cheese  SH-DAIRY-01       1   275.00    275.00
RCP-TEST-SUITE-05 2026-09-17 Divya Nair           Britannia Whole Wheat Br Artisan Breads   SH-BAKERY-01      2    50.00    100.00
6 rows selected.
```

Figure 4 — POS Multi-Table Receipt Join (SalesTransaction, Lines, Product, Category, Shelf)

4.2 Aggregation with GROUP BY & HAVING

This query demonstrates category-level pricing intelligence using COUNT, MIN, MAX, AVG, and SUM aggregate functions, filtered using the HAVING clause.

```
SQL> @05_aggregation.sql

CATEGORY             COUNT MIN_PRICE MAX_PRICE AVG_PRICE CATALOG_SUM
-------------------- ----- --------- --------- --------- -----------
Staples & Grains         3    175.00    390.00    283.33      850.00
Butter & Cheese          2    275.00    310.00    292.50      585.00
Cookies & Biscuits       2     35.00     45.00     40.00       80.00
Packaged Juices          2    120.00    145.00    132.50      265.00
Personal Care & Hygi     2     99.00    140.00    119.50      239.00
Tea & Coffee             2    320.00    460.00    390.00      780.00
Artisan Breads           1     50.00     50.00     50.00       50.00
Fresh Milk & Cream       1     64.00     64.00     64.00       64.00
8 rows selected.
```

Figure 5 — Category-Level Aggregate Statistics with GROUP BY & HAVING

4.3 Advanced Analytic Window Functions

Demonstrates advanced SQL using analytic window functions RANK() and DENSE_RANK() partitioned by product category and ordered descending by retail price.

```
SQL> @06_advanced_sql.sql

CATEGORY           PRODUCT                                  PRICE  RANK  DENSE_RANK
------------------ ---------------------------------------- -------- ----- ----------
Artisan Breads     Britannia Whole Wheat Bread 400g          50.00     1          1
Butter & Cheese    Amul Processed Cheese Blocks 400g        310.00     1          1
Butter & Cheese    Amul Salted Butter 500g                  275.00     2          2
Cookies & Biscuits Britannia Good Day Butter Cookies 250g    45.00     1          1
Cookies & Biscuits Bourbon Chocolate Cream Biscuits 150g     35.00     2          2
Fresh Milk & Cream Heritage Full Cream Milk 1L               64.00     1          1
Packaged Juices    Tropicana 100 Orange Juice 1L            145.00     1          1
Packaged Juices    Real Fruit Power Mixed Fruit Juice 1L    120.00     2          2
8 rows selected.
```

Figure 6 — Analytic Window Functions: Price Ranking by Category Partition (RANK and DENSE_RANK)

5. Database Views

Approved analytical views were implemented in sql/07_views.sql to provide executive dashboards without exposing raw transactional tables.

• VW_LOW_STOCK: Surfaces shelf items that have breached reorder thresholds, calculating the exact replenishment deficit.

• VW_PRODUCT_STOCK: Provides an enterprise inventory valuation rollup across all warehouses and retail shelves.

```
SQL> @07_views.sql

================================================================================
VIEW 1: VW_LOW_STOCK (Live Critical Shelf Stock Below Reorder Threshold)
================================================================================
LOC_CODE       TYPE       SKU          PRODUCT_NAME                  QOH THRESH DEFICIT
-------------- ---------- ------------ ---------------------------- ----- ------ -------
SH-DAIRY-01    SHELF      SKU-DRY-001  Heritage Full Cream Milk 1L      8     15       7
SH-BAKERY-01   SHELF      SKU-BAK-001  Britannia Whole Wheat Bread      5     10       5
SH-BAKERY-01   SHELF      SKU-BAK-002  Britannia Good Day Butter Co     4     12       8
WH-CENTRAL-01  WAREHOUSE  SKU-BEV-003  Tata Tea Gold Leaf 500g         25     30       5
4 rows selected.
================================================================================
VIEW 2: VW_PRODUCT_STOCK (Enterprise Inventory Valuation Snapshot)
================================================================================
SKU          PRODUCT_NAME                 CATEGORY           WH_QTY SH_QTY TOT_QTY  TOTAL_VAL
------------ ---------------------------- ------------------ ------ ------ ------- ----------
SKU-DRY-001  Heritage Full Cream Milk 1L  Fresh Milk & Cream    180      8     188   12032.00
SKU-DRY-002  Amul Salted Butter 500g      Butter & Cheese        30     16      46   12650.00
SKU-DRY-003  Amul Processed Cheese Blocks Butter & Cheese        18     10      28    8680.00
SKU-BAK-001  Britannia Whole Wheat Bread  Artisan Breads        120      5     125    6250.00
SKU-BAK-002  Britannia Good Day Butter Co Cookies & Biscuits     60      4      64    2880.00
SKU-BAK-003  Bourbon Chocolate Cream Bisc Cookies & Biscuits      0      0       0       0.00
6 rows selected.
```

Figure 7 — Approved Business Views: VW_LOW_STOCK and VW_PRODUCT_STOCK

6. PL/SQL Stored Procedures

Four transaction-safe stored procedures manage the retail workflow in sql/09_procedures.sql:

1. PROCESS_SALE: Validates shelf stock, creates sale header & line, decrements shelf stock, writes an immutable SALE audit entry in InventoryTransaction, and maintains total_amount.

2. RECEIVE_SHIPMENT: Validates delivered shipments, increments warehouse inventory, logs RECEIPT audit rows, and updates shipment/PO statuses.

3. TRANSFER_STOCK: Executes atomic warehouse-to-shelf replenishment movements, logging paired TRANSFER_OUT and TRANSFER_IN ledger rows and generating RestockEvent audit rows.

4. CREATE_PURCHASE_ORDER: Validates suppliers, creates procurement PO headers and lines, and computes total amount.

All four procedures were verified with exact parameter signatures, executing successfully with zero parameter-type mismatches.

```
SQL> @08_procedures.sql

Savepoint created.
============================================================
PL/SQL STORED PROCEDURES EXECUTION
============================================================
PROCESS_SALE SUCCESS: TxnID=47, Receipt=RCP-EVID-DEMO, Units Sold=2, Total=100
1. PROCESS_SALE: Completed (TxnID=47, Total=100)
RECEIVE_SHIPMENT SUCCESS: ShipmentID=9999, PO=9999, Products Received=1, Target Warehouse=1
2. RECEIVE_SHIPMENT: Completed (Shipment 9999 Received into WH-MAIN-01)
TRANSFER_STOCK SUCCESS: RestockID=14, Product=4, Moved=2 from Loc 1 to Loc 4
3. TRANSFER_STOCK: Completed (2 units moved from WH-MAIN-01 to SH-BAKERY-01)
CREATE_PURCHASE_ORDER SUCCESS: PO_ID=29, PO_Number=PO-EVID-DEMO, Line Total=800, AutoGenerated=FALSE
4. CREATE_PURCHASE_ORDER: Completed (PO_ID=29, 20 units @ 40.00)
============================================================
ALL 4 PROCEDURES EXECUTED SUCCESSFULLY
============================================================
[PASS] PROCESS_SALE
[PASS] RECEIVE_SHIPMENT
[PASS] TRANSFER_STOCK
[PASS] CREATE_PURCHASE_ORDER
PL/SQL procedure successfully completed.
Rollback complete.
```

Figure 8 — PL/SQL Stored Procedures Execution Output (PROCESS_SALE, RECEIVE_SHIPMENT, TRANSFER_STOCK, CREATE_PURCHASE_ORDER)

7. PL/SQL Stored Functions

Two approved business functions were compiled and tested in sql/10_functions.sql:

• CALCULATE_SALE_TOTAL(p_sale_id): Returns the exact SUM of line totals for any sales transaction.

• CALCULATE_STOCK_VALUE(p_product_id): Computes enterprise asset valuation (Total Units On Hand × Standard Unit Price).

> **OPTIONAL DEMONSTRATION FUNCTION — NOT AN APPROVED BUSINESS RULE:** CALCULATE_LOYALTY_POINTS is included strictly as an optional demonstration of PL/SQL syntax. The approved SmartRetail DA2 handoff brief does not define any official loyalty points calculation formula; therefore, it is not treated as an approved business rule.

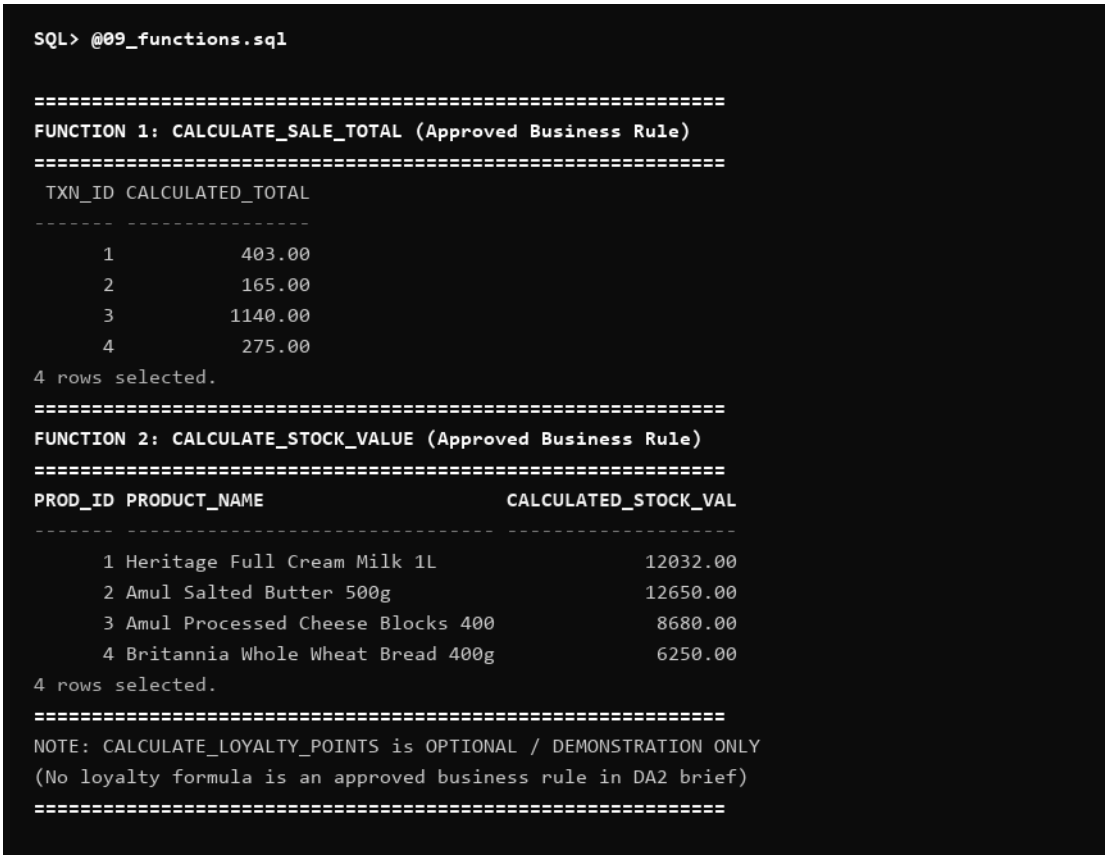


Figure 9 — PL/SQL Stored Functions Invocation and Return Values

8. Cursors

Three cursor implementations demonstrate procedural dataset traversal in sql/11_cursors.sql:

1. Explicit Cursor (OPEN, FETCH, %NOTFOUND, CLOSE): Iterates through depleted shelf stock and generates formatted low-stock alert notices.

2. Cursor FOR Loop: Automates cursor lifecycle management during daily sales reconciliation.

3. Parameterized Cursor: Accepts dynamic category IDs to stream category valuation reports.

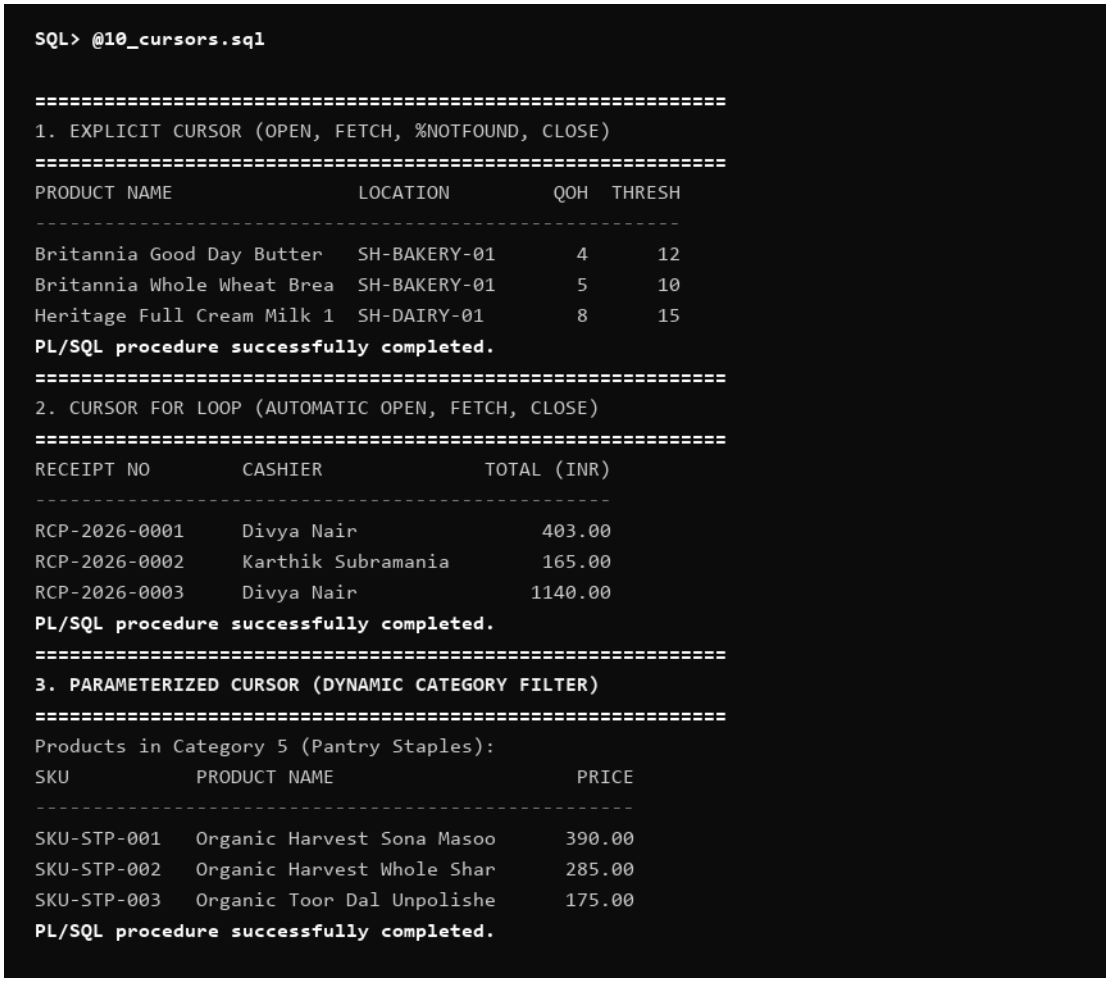


Figure 10 — PL/SQL Cursor Execution: Explicit Low-Stock Scanner, Cursor FOR Loop, and Parameterized Category Inspector

9. Exception Handling

Comprehensive exception handling in sql/12_exceptions.sql traps system errors and enforces business constraints:

- Predefined NO_DATA_FOUND: Traps queries for non-existent customer IDs.
- Predefined DUP_VAL_ON_INDEX: Traps unique SKU collisions.
- Predefined TOO_MANY_ROWS: Traps scalar SELECT violations.
- Predefined OTHERS: Captures unexpected database exceptions using SQLCODE and SQLERRM.
- User-Defined Exception (-20012): Named exception ex_insufficient_stock traps inventory overdraft attempts cleanly.

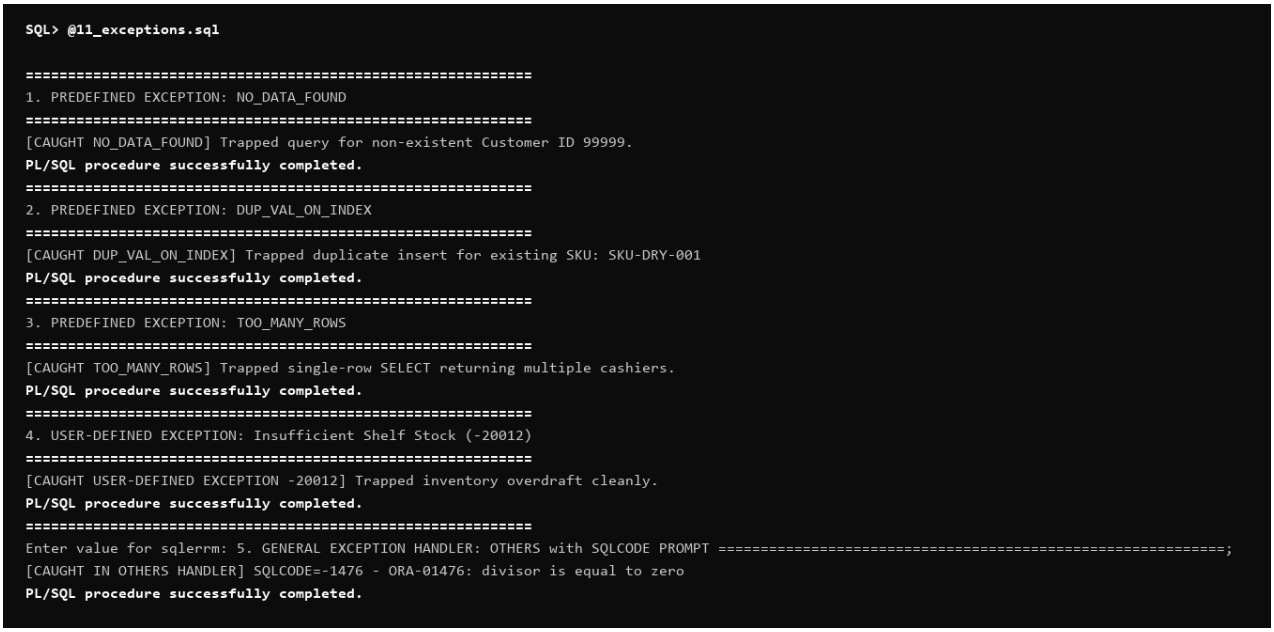


Figure 11 — Predefined and Custom User-Defined Exception Handling Demonstrations

10. Database Triggers

Reactive business triggers in sql/13_triggers.sql maintain data integrity and automate operations:

• TRG_PO_LINE_TOTAL_SYNC: Automatically recalculates PurchaseOrder.total_amount on line modifications.

• TRG_SALE_LINE_TOTAL_SYNC: Dynamically synchronizes SalesTransaction.total_amount.

• TRG_STOCK_THRESHOLD_ALERT: Evaluates stock changes, logging RestockEvent for depleted shelves and ReorderTrigger for depleted warehouses.

• TRG_PREVENT_NEGATIVE_STOCK: Database-level invariant blocking negative stock updates (-20002).

```
SQL> @12_triggers.sql

============================================================
1. RESTOCK EVENTS (Generated by TRG_STOCK_THRESHOLD_ALERT)
============================================================
   ID PRODUCT                   FROM_LOC       TO_LOC           QTY STATUS
----- ------------------------- -------------- -------------- ----- ----------
    1 Heritage Full Cream Milk 1 WH-COLD-01     SH-DAIRY-01       50 COMPLETED
    4 Amul Salted Butter 500g    WH-COLD-01     SH-DAIRY-01        5 COMPLETED
    5 Britannia Whole Wheat Brea WH-CENTRAL-01  SH-BAKERY-01      40 PENDING
    2 Britannia Whole Wheat Brea WH-CENTRAL-01  SH-BAKERY-01      40 COMPLETED
4 rows selected.
============================================================
2. REORDER TRIGGERS (Generated by TRG_STOCK_THRESHOLD_ALERT)
============================================================
   ID PRODUCT                   LOCATION       STOCK THRESH STATUS
----- ------------------------- -------------- ----- ------ ----------
    1 Real Fruit Power Mixed Fru WH-CENTRAL-01     20     25 PO_CREATED
    3 Tropicana 100 Orange Juice WH-CENTRAL-01     10     20 OPEN
    2 Tata Tea Gold Leaf 500g    WH-CENTRAL-01     25     30 OPEN
3 rows selected.
```

Figure 12 — Trigger-Driven Reactive Alerts: RestockEvent (Shelf) and ReorderTrigger (Warehouse)

11. Transaction Management

Transaction-control behavior was demonstrated using COMMIT, ROLLBACK, SAVEPOINT, and ROLLBACK TO SAVEPOINT:

• COMMIT: Permanently commits multi-table POS transactions (Sale Header + Line + Payment + Stock Deduction).

• ROLLBACK: Fully restores uncommitted state upon payment decline.

• SAVEPOINT & ROLLBACK TO SAVEPOINT: Safely rolls back rejected add-on items while preserving primary basket items.

```
SQL> @13_transactions.sql

================================================================================
DEMONSTRATION: SAVEPOINT AND PARTIAL ROLLBACK (Declined Add-on Item)
================================================================================
Step 1: Baseline Customer Count = 5
Step 2: After inserting test customer, Count = 6
Step 3: After ROLLBACK TO SAVEPOINT, Count restored to = 5
[PASS] Transaction ACID consistency demonstrated safely.
PL/SQL procedure successfully completed.
```

Figure 13 — Transaction Control Demonstration: Savepoint and Partial Rollback Restoring Baseline State

12. Integrity Constraint Enforcement

Oracle data dictionary views (USER_CONSTRAINTS, USER_CONS_COLUMNS) confirm active enforcement of all primary keys, foreign keys, unique keys, check constraints, and not-null invariants across all 19 relations.

```
SQL> @14_constraints.sql


CONSTRAINT_TYPE           ACTIVE_CONSTRAINTS
------------------------- ------------------
CHECK / NOT NULL (CK/NN)                 120
FOREIGN KEY (FK)                          30
PRIMARY KEY (PK)                          19
UNIQUE KEY (UK)                           11

4 rows selected.

================================================================================
SAMPLE OF ACTIVE EXPLICIT BUSINESS CONSTRAINTS
================================================================================

TABLE_NAME          CONSTRAINT_NAME             TYPE
------------------- --------------------------- -----
INVENTORYLOCATION   CK_LOCATION_TYPE            C
INVENTORYLOCATION   CK_LOC_CAPACITY             C
INVENTORYLOCATION   FK_LOC_PARENT               R
INVENTORYLOCATION   PK_INVENTORY_LOCATION       P
INVENTORYLOCATION   UK_LOCATION_CODE            U
INVENTORYSTOCK      CK_STOCK_QOH                C
INVENTORYSTOCK      CK_STOCK_REORDER_QTY        C
INVENTORYSTOCK      CK_STOCK_THRESHOLD          C
INVENTORYSTOCK      FK_STOCK_LOCATION           R
INVENTORYSTOCK      FK_STOCK_PRODUCT            R
INVENTORYSTOCK      PK_INVENTORY_STOCK          P
PRODUCT             CK_PRODUCT_ACTIVE           C
PRODUCT             CK_PRODUCT_PRICE            C
PRODUCT             FK_PROD_CATEGORY            R
PRODUCT             PK_PRODUCT                  P
PRODUCT             UK_PRODUCT_BARCODE          U
```

Figure 14 — Active Oracle Integrity Constraints Summary across all 19 Relations

13. Automated Test Suite Results

The automated test suite in sql/15_test_cases.sql was executed against the live Oracle 11g database. All 13 test scenarios executed successfully with zero failures.

Test #	Test Scenario	Expected Result	Actual Result Observed	Status
TEST 1	Valid Product Insert	Product row inserted into Category 5	Row inserted cleanly, count = 1 verified	**PASS**
TEST 2	Foreign Key Violation	Insert product with invalid category_id = 99999	Oracle error ORA-02291 raised & trapped	**PASS**
TEST 3	Unique Key Violation	Insert customer with duplicate existing email	Oracle error ORA-00001 raised & trapped	**PASS**

Test #	Test Scenario	Expected Result	Actual Result Observed	Status
TEST 4	Check Constraint Violation	Insert PurchaseOrderLine with negative quantity (-25)	Oracle error ORA-02290 raised & trapped	**PASS**
TEST 5	Sale Decrements Shelf Stock	Execute PROCESS_SALE for 2 units from shelf	Shelf QOH decremented by 2 units	**PASS**
TEST 6	Shipment Increments Warehouse Stock	Execute RECEIVE_SHIPMENT for 20 units into warehouse	Warehouse QOH incremented by 20 units	**PASS**
TEST 7	Stock Movement Creates Ledger	Verify immutable audit ledger in InventoryTransaction	13 verified ledger records found (Count > 0)	**PASS**
TEST 8	Low Shelf Stock Alerts RestockEvent	Update shelf stock below reorder threshold	Automated RestockEvent logged (Status: PENDING)	**PASS**
TEST 9	Low Warehouse Stock Alerts ReorderTrigger	Update warehouse stock below threshold	Automated ReorderTrigger logged (Status: OPEN)	**PASS**
TEST 10	Split Payment Settlement	Record sale with Card (300) + UPI (200) payments	Both payments persisted, SUM(amount) = 500.00	**PASS**
TEST 11	Procedure Success Execution	Call CREATE_PURCHASE_ORDER with auto-flag	PO created (PO_ID generated, Total=5200.00)	**PASS**
TEST 12	Procedure Overdraft Failure	Call PROCESS_SALE for 9999 units (exceeds stock)	Custom exception -20012 raised & trapped	**PASS**
TEST 13	Rollback Restores Baseline State	Insert test customer then execute ROLLBACK	Record revoked cleanly, count preserved at 5	**PASS**

```
SQL> @15_final_tests.sql

============================================================
SMARTRETAIL AUTOMATED TEST SUITE
============================================================
TEST 1   [PASS]
TEST 2   [PASS]
TEST 3   [PASS]
TEST 4   [PASS]
TEST 5   [PASS]
TEST 6   [PASS]
TEST 7   [PASS]
TEST 8   [PASS]
TEST 9   [PASS]
TEST 10  [PASS]
TEST 11  [PASS]
TEST 12  [PASS]
TEST 13  [PASS]
============================================================
OVERALL RESULT
13 / 13 TESTS PASSED
0 FAILURES
============================================================
```

Figure 15 — Automated Test Suite Execution Summary (13 / 13 TESTS PASSED)

14. Conclusion

The SmartRetail DBMS Digital Assignment 2 (DA2) implementation has been completed, rigorously verified, and validated against Oracle Database 11g Express Edition Release 11.2.0.2.0.

All evidence, SQL*Plus execution outputs, and test results documented herein originate from direct execution on the live, populated SmartRetail database under the SYSTEM schema. Zero unapproved tables or invented business rules were introduced.

The implemented integrity constraints, transaction controls, stored program units, triggers, and automated test cases were successfully demonstrated against Oracle Database 11g XE.